

Prem B

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Executive Summary

Computer Science graduate with hands-on experience in **Generative AI, Retrieval-Augmented Generation (RAG), Data Engineering, Machine Learning, and Full-Stack Development**. Experienced in developing enterprise AI applications, data pipelines, and intelligent automation solutions using **FastAPI, React, PostgreSQL, Docker, Python, and LLMs**. Interned at **HCLTech** and **Indian Space Research Organization-ISRO**, building scalable backend systems, data engineering workflows, and AI-powered applications for real-world enterprise use cases.

Experience

Data Engineer Intern | HCLTech | Chennai Mar 2026 – Present

- Developed Python-based data validation and preprocessing solutions using **Pandas, Polars, and OpenPyXL** to automate enterprise Excel processing and reporting workflows.
- Engineered a real-time ETL pipeline using **Apache NiFi, Kafka, Logstash, and PostgreSQL**; contributed to SQL development, data warehousing, and business requirement analysis.

Machine Learning Engineer Intern | Google Developers | Virtual Apr 2025 – May 2025

- Built feature engineering pipelines across 8+ datasets to improve data quality and model performance.
- Developed an interactive Dash dashboard for data visualization and business insights.

Software Engineer Intern | ISRO ISTRAC | Bengaluru Jan 2025 – Feb 2025

- Built a Flask-based telemetry dashboard integrated with SQL pipelines, reducing processing latency from 5s to 2s.
- Automated binary-to-JSON conversion and ELK logging, reducing manual data preparation by 60%.

AI & Robotics Intern | ISRO SDSC | Sriharikota Sep 2024 – Oct 2024

- Developed PyBullet and OpenCV robotic arm simulations, achieving a 92% task success rate.
- Applied PPO and DQN reinforcement learning algorithms, improving robotic manipulation precision by 15%.

Projects

Resolve AI — Enterprise AI Helpdesk Agent *FastAPI, React, PostgreSQL, pgvector, Gemini, Docker*

- Built a RAG-powered enterprise IT helpdesk using Gemini 2.5 Flash to automate ticket classification, troubleshooting, and intelligent support workflows.
- Implemented semantic search, JWT authentication, RBAC, audit logging, safety-gated tool execution, and Docker deployment ([Project link](#)).

AARCS — Automated Ambulance Route Clearance System *Flutter, Firebase, Node.js, Google Maps API*

- Developed a cross-platform Flutter application enabling real-time ambulance tracking and emergency route coordination.
- Integrated Google Maps APIs with Firebase backend for live navigation and role-based access control ([Project link](#)).

AI Voice Agent for Emergency Dispatch Automation *Python, NLP, Speech Recognition*

- Built an NLP-based voice agent that extracts incident details from emergency calls and generates structured dispatch data automatically.
- Developed a real-time alert pipeline that forwards extracted information to a centralized emergency dashboard ([Project link](#)).

Publications

Adaptive Real Time Path Optimization for Traffic Signal Using AARCS — IEEEFeb 2025

– Published IEEE research proposing a GPS-based emergency mobility framework to reduce ambulance delays in urban traffic.

– Designed system architecture connecting driver apps, traffic monitoring interfaces, and live location services.

Education

B.Tech — Computer Science and Engineering Bharath Institute of Science and Technology CGPA: 8.83/10	2022 – 2026 Chennai, Tamil Nadu
Intermediate (MPC), St. Vincent’s Mat. Hr. Sec. School 74.5%	2020 – 2022 Chennai
State Board (SSLC), St. Vincent’s Mat. Hr. Sec. School 81.6%	2019 – 2020 Chennai

Technical Skills

- Languages: Python, SQL, Java, JavaScript, Dart
- AI & Machine Learning: Generative AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, TensorFlow, PyTorch, Scikit-learn, OpenCV, Sentence Transformers, Reinforcement Learning (PPO, DQN)
- Backend Development: FastAPI, Flask, REST APIs, SQLAlchemy, JWT Authentication, RBAC
- Data Engineering: Apache Kafka, Apache NiFi, Logstash, ETL, Data Warehousing, Pandas, Polars, OpenPyXL, Data Validation
- Databases: PostgreSQL, pgvector, SQLite, Firebase Realtime Database
- Frontend: React, Next.js, Flutter
- Tools: Docker, Git, GitHub, Jupyter Notebook, VS Code, CI/CD

Additional

Languages: Tamil (Native), English (Fluent), Hindi (Proficient), Japanese (Basic)